International Monetary Fund

Singapore Regional Training Institute

OT26.07 Cohort Training in Nowcasting

September 21-October 2, 2026

Workshop 2 (Part 1): Useful Statistical Procedures for Nowcasting, Part 1: Time Series Analysis

Interpolating time-series to higher frequencies using the Chow-Lin procedure in EViews: Tonga's Real GDP

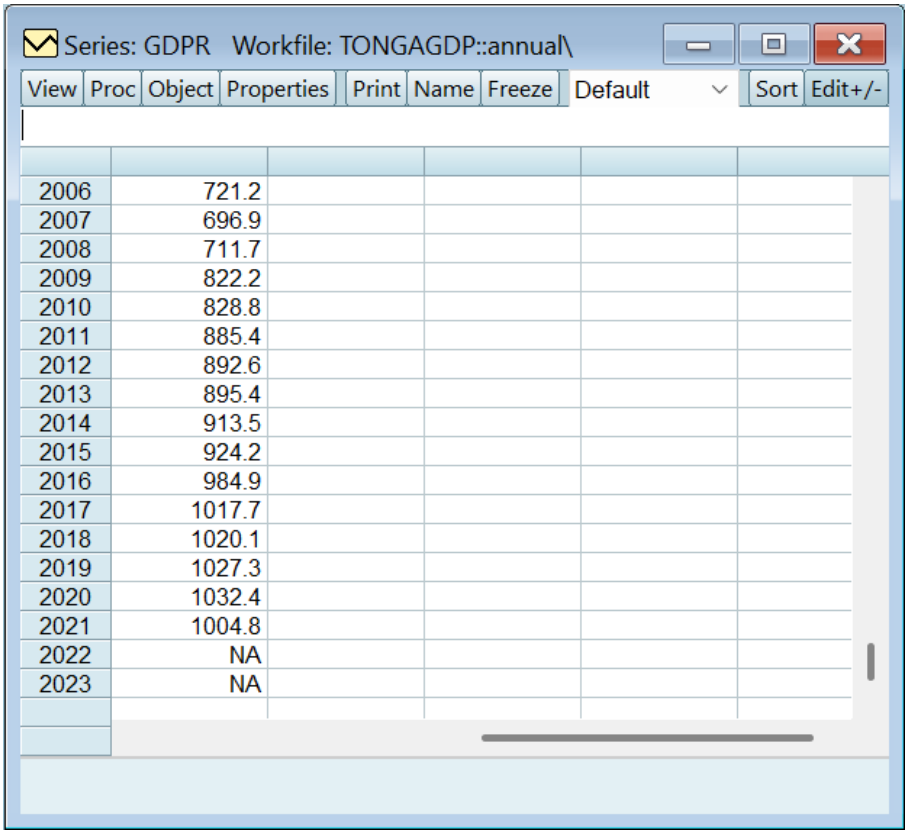
The **Chow–Lin (1971)** procedure is one of the most widely used statistical methods for **temporal disaggregation**—that is, converting a **low-frequency series** (e.g., annual GDP) into a **high-frequency series** (e.g., semi-annual or quarterly GDP) using one or more related indicators available at the higher frequency. Its popularity comes from its **simplicity**, **statistical grounding**, and **flexibility** in handling autocorrelation structures in the underlying high-frequency residuals.

This is a guide to the Chow-Lin procedure available in EViews. It provides step-by-step instructions on how to interpolate a time-series to a higher frequency using (when available) higher-frequency indicators to estimate the missing data points. The result is a series with more non-missing observations than the original series.

We illustrate the Chow-Lin procedure using the real GDP data of Tonga, which is available at an annual frequency for 1994-2021 in the sample workfile (**tongagdp.wf1**). The aim is to interpolate real GDP to the semi-annual frequency (i.e., two observations per year).

Figure 1 shows the EViews spreadsheet view of Tonga's real GDP data for 2003-2021:

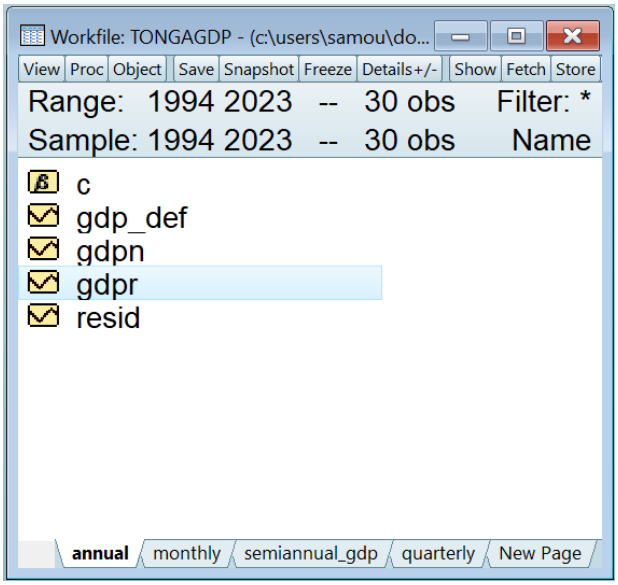
Figure 1: EViews' Spreadsheet View of Tonga's Real GDP



2006	721.2
2007	696.9
2008	711.7
2009	822.2
2010	828.8
2011	885.4
2012	892.6
2013	895.4
2014	913.5
2015	924.2
2016	984.9
2017	1017.7
2018	1020.1
2019	1027.3
2020	1032.4
2021	1004.8
2022	NA
2023	NA

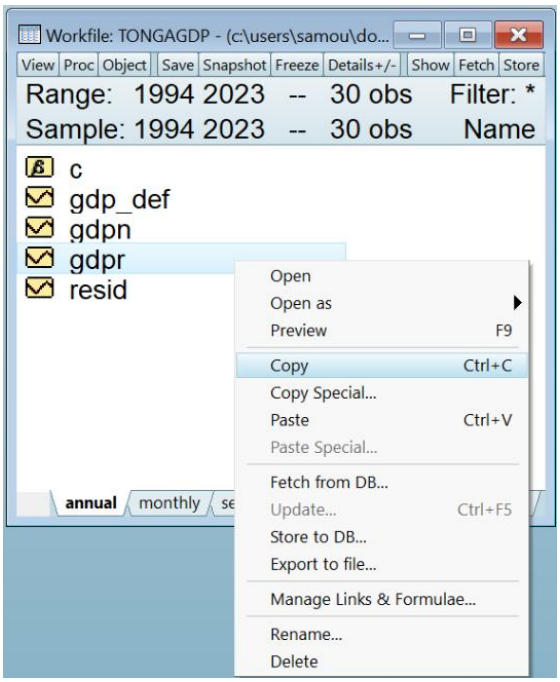
To better understand the interpolation problem, we will first move annual real GDP to the semi-annual frequency (2 observations per year) without replacing the resulting missing values.

To do so, open the workfile containing Tonga's real GDP data and click on the annual pagefile:

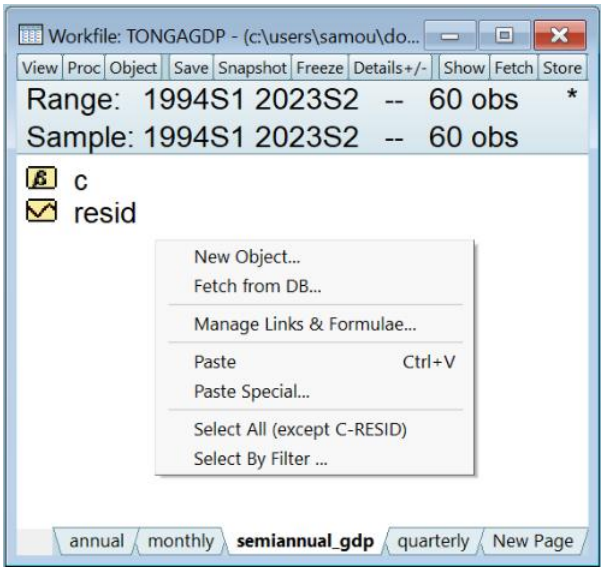


Note that Tonga's real GDP is highlighted. It is named "**gdpr**" in the EViews workfile and has been selected by clicking on the name.

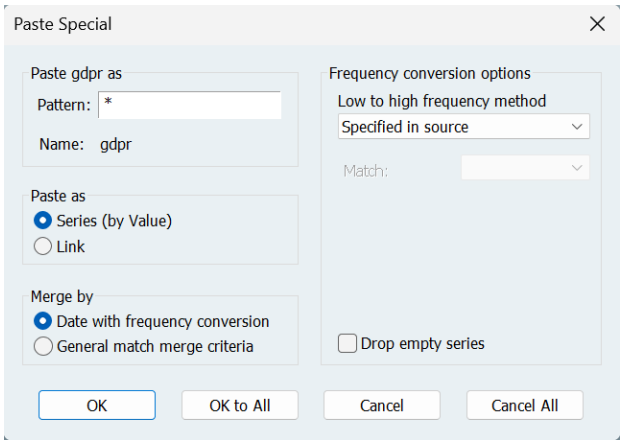
Position the mouse-pointer over **gdpr** and then right-click and invoke "**Copy (Ctrl+C)**":



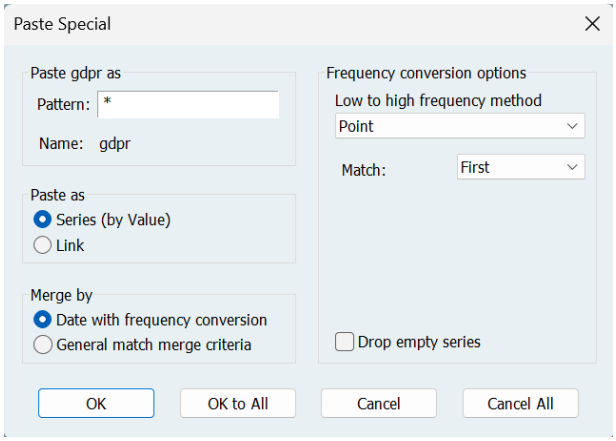
After invoking the copy command, move to the semi-annual page file ("**semiannual_gdp**"), right-click anywhere on that page and invoke "**Paste Special**":



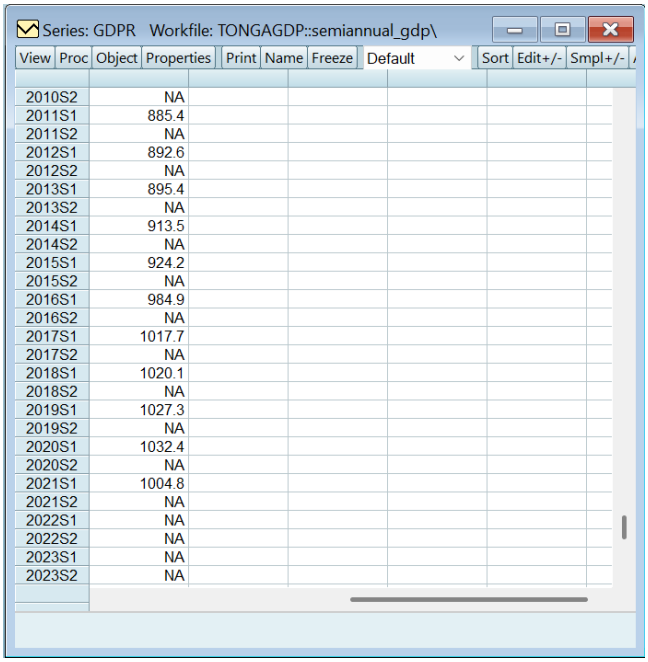
Doing so brings up the following dialog box:



From the "**Low to high frequency method**" select "**Point**" and "**First**" for "**Match**" (to be consistent with Tonga's financial year (July-June)):



Now click **OK** to move the real GDP series to the semi-annual page file without any interpolation:



2010S2	NA
2011S1	885.4
2011S2	NA
2012S1	892.6
2012S2	NA
2013S1	895.4
2013S2	NA
2014S1	913.5
2014S2	NA
2015S1	924.2
2015S2	NA
2016S1	984.9
2016S2	NA
2017S1	1017.7
2017S2	NA
2018S1	1020.1
2018S2	NA
2019S1	1027.3
2019S2	NA
2020S1	1032.4
2020S2	NA
2021S1	1004.8
2021S2	NA
2022S1	NA
2022S2	NA
2023S1	NA
2023S2	NA

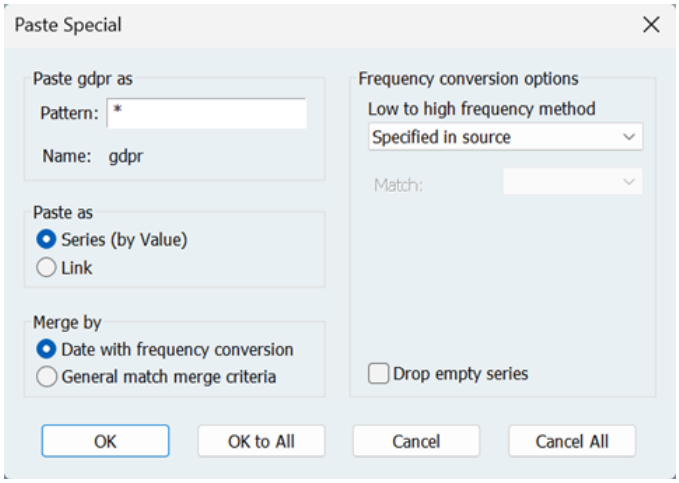
Notice that the values of Tonga's annual real GDP have been migrated to a semi-annual timeframe. The "point" procedure introduces a missing value (NA) for every second period ("S2") of each financial year. Moreover, the data is positioned in the first half of the calendar year, which is appropriate given that Tonga's annual real GDP figure measures output for the July-June financial year.

The role of the Chow-Lin procedure is to replace these missing values with estimated values. It is a regression-based[1] interpolation technique that estimates the missing values of a series by relating one or more higher-frequency indicator series to a lower-frequency benchmark series. It can also estimate the missing values without auxiliary indicators.

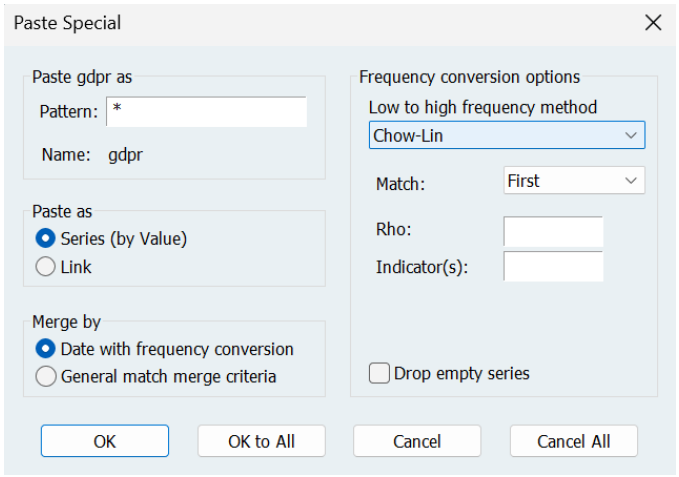
[1] A brief review of the regression based estimator underlying the Chow-Lin procedure is provided in Appendix A.

We first demonstrate the procedure initially without using indicators:

Step 1: Return to the annual pagefile and copy the annual GDP series "**gdpr**". Then move to the "**semiannual_gdp**" pagefile, right-click, and invoke "**Paste Special**". As before, the following dialog box appears:

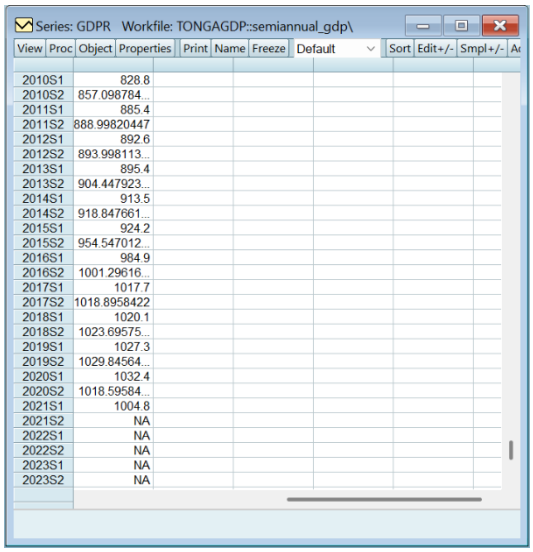


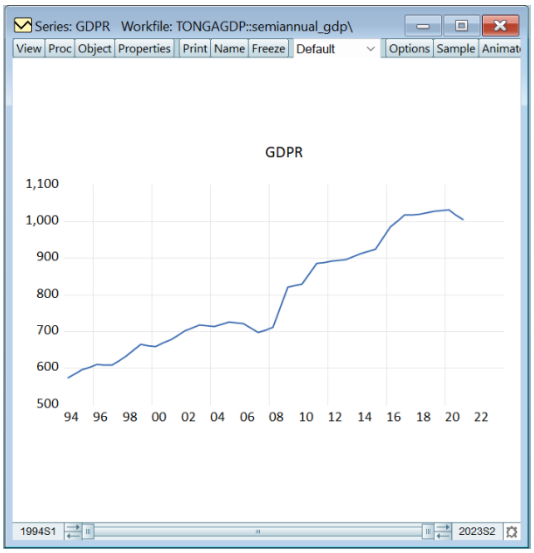
Now select the "**Chow-Lin**" option available in the "**Low to high frequency method**:



Note that "Rho:" in the above dialog box reflects the persistence in the error-term of the regression equation used to predict the unknow values. Lower values of ρ tend to introduce more noise into the interpolated series. The default value for ρ is 0.9, but it can be set to any value between 0 and 1.

Click "**OK**" to interpolate annual real GDP to the semi-annual frequency. Given the use of "*" in the dialog box labelled with "**Paste gdpr as**" the result will be in the series named "**gdpr**" in the "**semiannual_gdp"** pagefile:

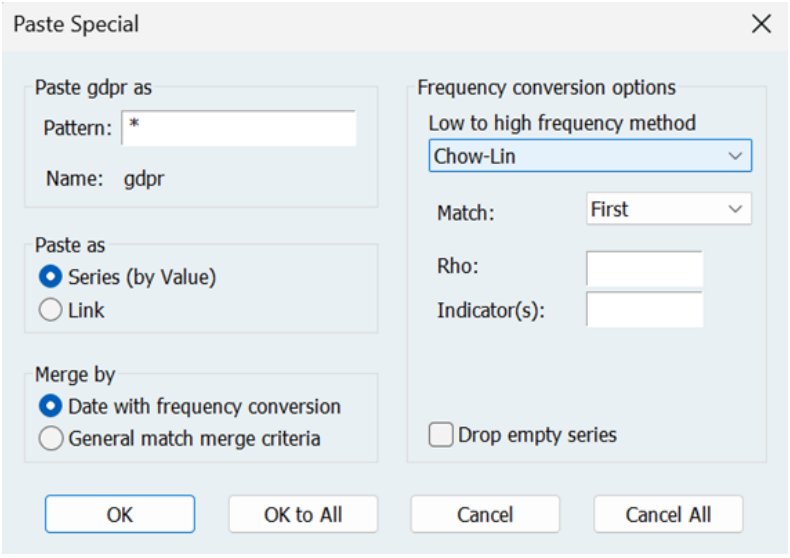




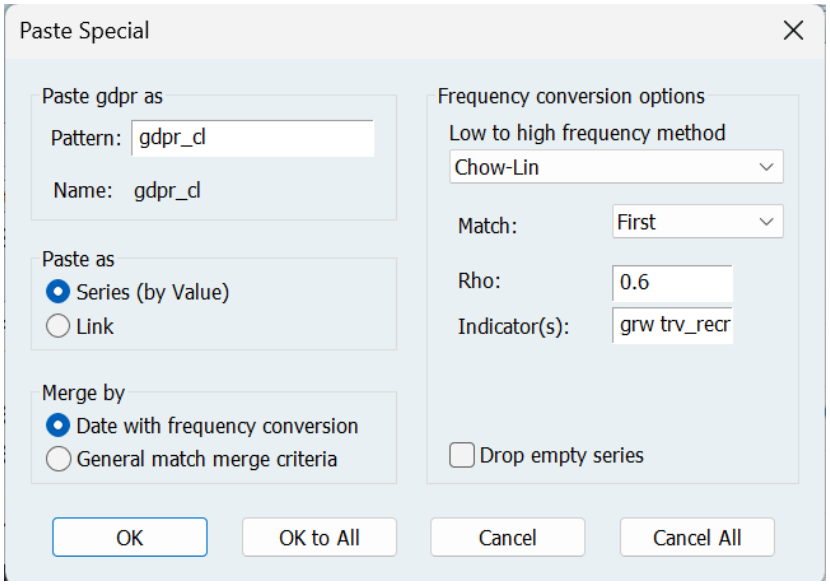
Note that the missing values in the original series have been replaced with estimates using the Chow-Lin procedure (without indicators). The known actual values of real GDP are not affected.

The Chow-Lin procedure can also account for higher frequency indicators to estimate the missing values. This is the preferred approach because the interpolation will account for any economic conditions embedded in the chosen indicators.

To demonstrate the application of the Chow-Lin with indicators, return to the following dialog-box:

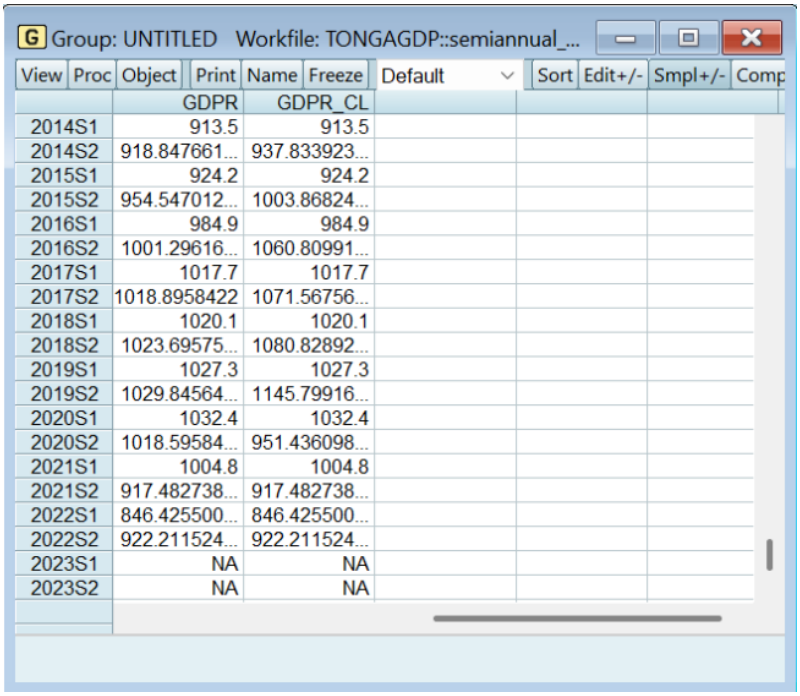


and populate the "**Indicator(s)**" text box with the name(s) of semi-annual series that may help predict the missing values in real GDP when moved to the semi-annual time frame. Here we use semi-annual indicators for credit growth ("**crd_grw**") and travel receipts ("**trv_recr**") in Tonga to do so. We also reduce the value of ρ to 0.6, thereby assuming less persistence in error term.[2]

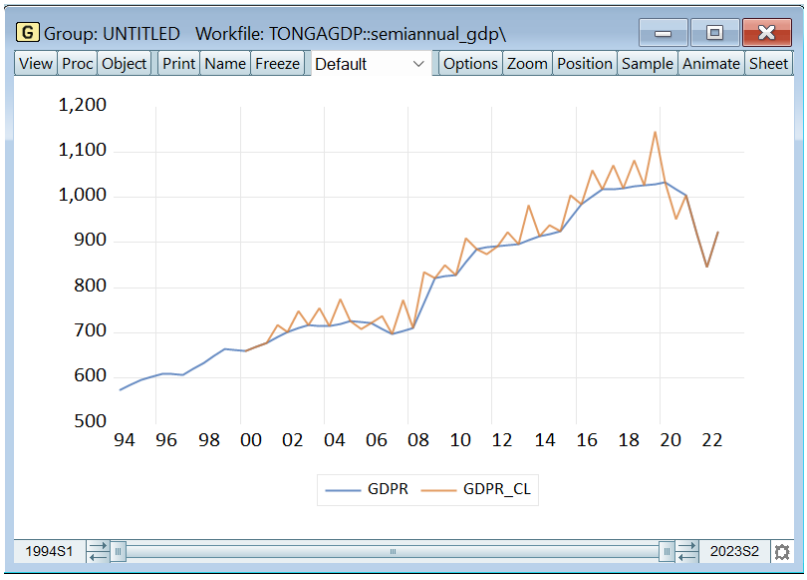


[2] The value of ρ will be estimated by EViews in this case.

The resulting series will be saved in a series called "**gdpr_cl**". The result for the original **gdpr** (Chow-Lin without any indicators) and **gdpr_cl** (Chow-Lin with indicators) is:



	GDPR	GDPR_CL
2014S1	913.5	913.5
2014S2	918.847661...	937.833923...
2015S1	924.2	924.2
2015S2	954.547012...	1003.86824...
2016S1	984.9	984.9
2016S2	1001.29616...	1060.80991...
2017S1	1017.7	1017.7
2017S2	1018.8958422	1071.56756...
2018S1	1020.1	1020.1
2018S2	1023.69575...	1080.82892...
2019S1	1027.3	1027.3
2019S2	1029.84564...	1145.79916...
2020S1	1032.4	1032.4
2020S2	1018.59584...	951.436098...
2021S1	1004.8	1004.8
2021S2	917.482738...	917.482738...
2022S1	846.425500...	846.425500...
2022S2	922.211524...	922.211524...
2023S1	NA	NA
2023S2	NA	NA



Lastly, the EViews' command to invoke the Chow-Lin procedure from its command window is:

```
copy(rho=0.6, c=chowlinf, overwrite) annual\GDPR gdpr_cl @indicator "crd_grw trv_recr"
```

Appendix: Econometrics underlying the Chow-Lin Procedure

The Chow–Lin procedure assumes an underlying **high-frequency** regression:

$$y = X\beta + u$$

where:

- y is the unobserved high-frequency target series
- X is the high-frequency indicator matrix
- $u \sim N(0, \sigma^2 V)$ with V typically an AR(1) covariance structure:

$$u_t = \rho u_{t-1} + \varepsilon_t$$

But we **do not observe** y. Instead, we observe the aggregated series:

$$Y = Cy$$

where C is the aggregation matrix (e.g., sums every 4 quarters into one annual value).

Chow–Lin uses **Generalized Least Squares (GLS)** on the aggregated model:

$$Y = CX\beta + Cu$$

The covariance of the aggregated residuals is:

$$\Omega = CVC'$$

The GLS estimator is:

$$\hat{\beta}(\rho) = (X'C'\Omega^{-1}CX)^{-1}X'C'\Omega^{-1}Y$$

But ρ is unknown. Chow–Lin estimates it by **concentrated maximum likelihood** or by minimizing the GLS residual sum of squares.

Once $\hat{\rho}$ and $\hat{\beta}$ are obtained, the high-frequency estimate is:

$$\hat{y} = X\hat{\beta} + \hat{u}$$

where the high-frequency residuals are recovered via:

$$\hat{u} = VC'\Omega^{-1}(Y - CX\hat{\beta})$$

This is the **best linear unbiased estimator (BLUE)** of the disaggregated series under the assumed AR(1) structure.